CSC 428
Lab 4
Points: 100 + 20 Bonus

Lab Overview

In cybersecurity and system monitoring, log files contain valuable information about system activities. However, to analyze logs using machine learning algorithms, we need to convert the text-based logs into numerical representations. This process is called "vectorization."

In this lab, you will create a program that:

1. Reads system log files
2. Identifies them as either normal or attack logs
3. Transforms the text into numerical features
4. Prepares a dataset ready for supervised machine learning
5. BONUS (20 Points): Applies a J48-C45 Decision Tree algorithm with 10-fold cross-validation

Learning Objectives

- Understand how text data is converted to numerical vectors for machine learning
- Implement a basic feature extraction pipeline for text data
- Prepare labeled datasets for supervised learning
- Gain practical experience with the scikit-learn library's text processing tools

Example Data

Below is a sample of the log file format you'll be working with:

```
"Time of Day","Process Name","PID","Operation","Path"
"3:11:17.8988718 PM","Explorer.EXE","2368","RegQueryV
"3:11:17.8988843 PM","Explorer.EXE","2368","RegSetVal
"3:11:17.8988899 PM","Explorer.EXE","2368","RegSetVal
"3:11:17.9011089 PM","SearchIndexer.exe","2700","File
"3:11:17.9011173 PM","SearchIndexer.exe","2700","File
"3:11:17.9011237 PM","SearchIndexer.exe","2700","File
"3:11:18.0114931 PM","VMwareTray.exe","2520","CreateF
```

Lab Requirements/Steps

1. Data Collection

- Download the log-files.zip from Canvas
- The dataset contains multiple CSV files
- Files with "Good" in the name represent normal system activity
- Files without "Good" in the name represent attack/suspicious activity

2. Data Preparation

- Read all CSV files from the dataset
- Assign labels: 1 for normal activity (Good), -1 for attack activity
- Clean the text by removing commas, quotation marks, and other characters that might interfere with vectorization
- Combine all log entries into a single dataset while preserving their labels

3. Vectorization

- Implement vectorization using scikit-learn's CountVectorizer
- Configure the vectorizer to:
  - Remove English stop words
  - Limit to the top 1000 most frequent terms
- Convert all log text into a feature matrix where:
  - Each row represents one log file
  - Each column represents the frequency of a specific term
  - The values show how many times each term appears in each log

4. Output Preparation

- Create a pandas DataFrame from the vectorized data
- Use the feature names (terms) as column headers
- Use the labels (1 or -1) as the index
- Save the resulting dataset as a CSV file

Implementation Guidelines

If you want to write the program using Python:

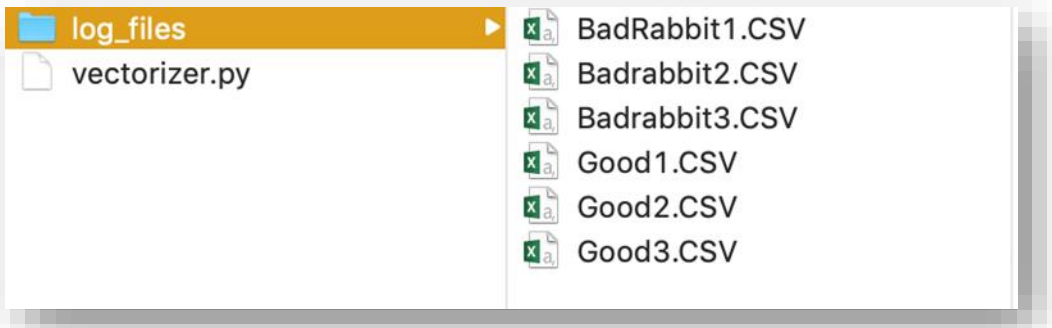
- Import the required libraries

```
import os
import pandas as pd
from sklearn.feature_extraction.text import CountVectorizer
```

- Define the parameter to store the path for script to read data. And define the parameters to store the labels and text to be vectorized.
  - The name of the files will tell you whether it belongs to a normal activity or not.

```
path = 'log_files'

labels = []
text = []
```



- Read the content from each file and create labels for them

```
for filename in os.listdir(path):
    if "Good" in filename:
        labels.append("1")
    else:
        labels.append("-1")
    filename = os.path.join(path, filename)
    print (filename)
    with open(filename,encoding='utf-8') as f:
        content = f.read()
    content.replace(",", " ")
    content.replace('"', " ")
    text.append(content)
```

- If the log file belongs to a normal activity, "1" will be assigned to it as a label. Otherwise, "-1" will be assigned to indicate it belongs to an attack activity.
- In order to perform vectorization, some characters such as commas and quotation mark need to be removed.

- Use CountVectorizer() to create a vectorizer for the text of log files. After the vectorization, you will be able to get the features and feature names of the content.

```
vectorizer = CountVectorizer(stop_words='english', max_features=1000)
```

    - stop_words='english' indicates that all the stop words in the content will be removed.
    - max_features=1000 indicates that 1000 features will be generated based on the frequency order of the terms in text.
- Save the features to a csv file by using pandas DataFrame.to_csv() function.
    - You may first need to covert the results to a dataframe
        - [DataFrame is a 2-dimensional labeled data structure with columns of potentially different types. You can think of it like a spreadsheet or SQL table. It is generally the most commonly used pandas object.]
    - The index of dataframe should be the labels.
    - Vectorizer can give you the names of the features, and you can use them as the column names for the dataframe.

BONUS (20 Points): Machine Learning Classification

Apply a J48 Decision Tree classifier with 10-fold cross-validation:

- Use your vectorized data as input features
- Implement the decision tree classifier with entropy criterion (equivalent to J48/C4.5)
- Apply 10-fold cross-validation to evaluate performance
- Calculate and report accuracy, precision, recall, and F1-score
- Analyze which features (terms) are most important for classification

Deliverables

You should submit:

1. Source Code: The complete Python (or other language) source code that implements the requirements above.
2. Lab Report: A PDF file named "**LastnameFirst-CSC428-Lab4.pdf**" that includes:

    a) The steps you processed data
    b) The necessary code snippet of your vectorizer
    c) Screenshots of the results
    d) If you complete the bonus section, include your decision tree results and analysis

Grading Criteria

- Basic Assignment (100 points):
  - Correct file reading and label assignment (25 points)
  - Text cleaning and preparation (25 points)
  - Proper vectorization implementation (25 points)
  - Correct DataFrame creation and CSV output (25 points)
- Bonus Assignment (20 points):
  - J48/C4.5 implementation with 10-fold cross-validation (10 points)
  - Performance metrics and feature importance analysis (10 points)